

25. A plane flies through Canada from West to East where the Earth's magnetic field is inclined at an angle below the horizontal, into the Earth.

Which of the diagrams shows the correct accumulation of charges on the plane after some time has passed?



A



B



C



D



Ans: B

Consider the vertical component of the Earth's magnetic field (which is perpendicular to the wing)

Apply Fleming's LHR to show that the force acting on +ve particles is towards the left wing, so the force acting on -ve particles will point towards the right wing. Hence the right wing is negative and the left wing is positive.

26 A sinusoidal alternating supply is connected to a $10\ \Omega$ resistor in series. The graph